

ITI TO SYLLABUS

Syllabus Units

1. Install and setup operating system and related software in a computer following safety precautions.
2. Create, format and edit document using word processing application software.
3. Create, format, edit and develop a workbook by using spreadsheet application software.
4. Create and customize slides for presentation.
5. Create and manage database file using MySQL.
6. Install, setup/configure, troubleshoot and secure computer network including Internet.
7. Develop web pages using HTML and word press.
8. Develop web pages using Java Script.
9. Create workbooks with advanced formulae, macros, charts, pivot tables and demonstrate ability to use Power tools.
10. Browse, select and transact using E commerce websites.
11. Secure information from Internet by using cyber security concept.
12. Explain Cloud concepts & services.
13. Write programs using Python / Java language.

Syllabus Unit Details

1. **Install and setup operating system and related software in a computer following safety precautions.**
 - Identify basic first aid and use them under different circumstances.
 - Identify different fire extinguisher and use the same as per requirement.
 - Assemble a computer.
 - Install and configure Windows OS.
 - Install the printer and other peripheral devices.
 - Install application software.
 - Troubleshoot the PC.
 - Execute DOS and LINUX commands.
 - Customize Windows and LINUX OS settings.

2. Create, format and edit document using word processing application software.

- Create your resume using editing/formatting options in a document.
- Create purchase order using tables and images.
- Create magazine using columns page borders, header footers.
- Create an invitation letter using mail merge for n invitees.

3. Create, format, edit and develop a workbook by using spreadsheet application software.

- Identify Excel tools in the Ribbon.
- Create mark sheet using a spreadsheet with data validation.
- Create a chart for the mark sheet.
- Create Pay slip using functions and formulae with sharing two different sheets/files.
- Create a table and Perform Sorting; filtering, Subtotal, validation and goal seek on a table.
- Prepare a pivot table on any existing table with data.
- Create a table and Perform Sorting; filtering, Subtotal, validation & goal seek on a table.
- Prepare a pivot table on any existing table with data.

4. Create and customize slides for presentation.

- Create simple presentations.
- Create presentations with tables, images & graphic elements.
- Create presentations with audio & video elements with transitions.

5. Create and manage database file by using MySQL.

- Create simple database on Relational Database in MySQL using data validation, filters, sorting, query.
- Import, Export, Link, Backup and Retrieve database in MySQL.
- Create query with functions, joins, sub-query.

9. Create workbooks with advanced formulas, macros, charts, pivot tables and demonstrate ability to use Power tools.

- Create workbooks with advanced functionalities in Excel.
- Create advanced charts & Pivot Tables.
- Create output files using specific Power tool.

10. Browse, select and transact using E-commerce websites.

- Place order for products from E commerce websites for purchase.
- Upload a product in E Commerce site for sale.
- Identify security issues in E- commerce and payment operations.

11. Secure information from Internet by using cyber security concept.

- Provide firewall security for Internet connection and Network System.
- Make backup copies of important file, data and information.
- Secure your Wi-Fi networks using wireless security features.

12. Explain Cloud concepts

- Create cloud concepts.
- Use common cloud services such as Office 365, Google Drive, Dropbox.
- Identify the phases of Application Development Life Cycle.
- Describe Roles in each of phases of the Application Development Life Cycle.

13. Write programs using Python/Java language.

- Install Python / Java.
- Perform operations on Python / Java; construct simple code and document these.
- Perform Document code segments using comments and documentation strings.
- Perform operations using in-built modules / libraries.

6. Install, setup/configure, troubleshoot & secure computer network including Internet.

- Identify different cables and connectors used in networking.
- Assign Computer Name and workgroup to a computer Prepare UTP cross cable & connect computers.
- Share a printer with Network.
- Share Internet using Windows Tools.
- Check Network connectivity.
- Configure HUB & Switch.
- Configure DHCP and firewall.
- Secure Network with various tools.
- Create E Mail ID and sending / receiving mails Perform text chat and video chat using social network sites Configure Outlook Express.

7. Develop web pages using HTML, CSS and Word press.

- Create Text, Lists, Tables, and Frames with HTML.
- Create Hyperlinks, Images and Multimedia Working with Forms and controls.
- Create Lists and Tables with CSS.
- Create Box Model by using borders, Padding, and Margin with CSS.
- Create CSS document by Grouping, Dimension, Display, Positioning, Floating, Align, Pseudo class, Navigation Bar, Image Sprites, Attribute selector).
- Create simple static Web Pages using internal styles (CSS) and external style.

8. Develop web pages using Java Script.

- Design a dynamic Web Page in JavaScript using various operators.
- Design a dynamic Web Page in JavaScript using various control statements & looping structures.
- Design a dynamic Web Page in JavaScript using strings and functions.
- Design a dynamic Web Page in JavaScript using Arrays and objects.
- Design a dynamic Web Page in JavaScript using Web Forms and images.